

FURTHER MATHEMATICS CURRICULUM

Complete Syllabi Specification for Ordinary Level (O'Level) & Advanced Level (A'Level)

1. ORDINARY LEVEL (O'LEVEL) FURTHER MATHEMATICS

1.1 PURE MATHEMATICS

Roots of Quadratic Equations

- Sum and product of roots for quadratic equations
- Forming quadratic equations with given specified roots
- Symmetric functions of roots ($\alpha^2 + \beta^2$, $\alpha^3 + \beta^3$, etc.)

Polynomials & Rational Functions

- Remainder Theorem and Factor Theorem application
- Synthetic division and polynomial long division
- Factorization of cubic and quartic polynomials
- Partial fractions (distinct linear, repeated linear, and irreducible quadratic denominators)

Indices, Logarithms & Surds

- Advanced laws of indices and complex logarithmic equations
- Exponential equations and change of base formulas
- Operations on surds and rationalization of compound denominators

Functions & Transformations

- One-to-one, many-to-one functions, determination of domain and range
- Composite functions $f(g(x))$ and inverse functions $f^{-1}(x)$
- Graph transformations: $f(x \pm a)$, $a f(x)$, $f(ax)$, $|f(x)|$

Inequalities & Absolute Values

- Quadratic inequalities and sign diagrams
- Rational inequalities involving algebraic fractions
- Absolute value equations and non-linear inequalities

Series & Sequences

- Arithmetic Progressions (AP) and Geometric Progressions (GP)
- Sum to infinity of convergent geometric series
- Sigma notation Σ and properties of summations
- Binomial Expansion for positive integral exponents $(a + b)^n$

Trigonometry

- Radian measure, arc length, and sector area calculations
- Trigonometric ratios for any angle using the unit circle
- Compound angle formulas: $\sin(A \pm B)$, $\cos(A \pm B)$, $\tan(A \pm B)$
- Double angle formulas: $\sin(2A)$, $\cos(2A)$, $\tan(2A)$
- Solving trigonometric equations within specified intervals

Calculus: Differentiation

- First principles derivative concept and limits
- Derivatives of x^n , trigonometric functions, e^x , and $\ln x$
- Product Rule, Quotient Rule, and Chain Rule
- Tangents and normals to algebraic and trigonometric curves
- Stationary points, maxima, minima, and points of inflection
- Rates of change and kinematics applications

Calculus: Integration

- Indefinite integration as reverse differentiation
- Standard integrals of algebraic, exponential, and trigonometric functions
- Definite integrals and fundamental theorem of calculus
- Integration by algebraic substitution
- Area under a curve and area between intersecting curves
- Volume of revolution about the x-axis and y-axis

Coordinate Geometry & Vectors

- Midpoint, distance, and gradient of straight lines
- Equation of a circle: $(x-a)^2 + (y-b)^2 = r^2$ and general form
- Intersection of lines and circles, tangents, and normals
- Vectors in 2D ($\mathbf{i}$, $\mathbf{j}$ notation), magnitude, direction, and dot product

1.2 MECHANICS & APPLIED MATHEMATICS

Kinematics

- Linear motion with constant acceleration (SUVAT equations)
- Motion graphs analysis ($s-t$ and $v-t$ graphs)
- Variable acceleration using differentiation and integration

Statics & Forces

- Resolution of forces and friction laws ($F \leq \mu R$)
- Equilibrium of a particle under coplanar forces
- Lami's Theorem and triangle of forces
- Moments, couples, and center of gravity of simple uniform bodies

Dynamics

- Newton's Laws of Motion and system application
- Connected particles (pulleys, inclined planes, towbars)
- Momentum, impulse, and conservation of linear momentum
- Work, kinetic energy, potential energy, power, and Work-Energy Principle

1.3 STATISTICS & PROBABILITY

Data Analysis & Summary Measures

- Mean, variance, and standard deviation for grouped and ungrouped data
- Quartiles, percentiles, and interquartile range (IQR)
- Box plots, cumulative frequency graphs, and stem-and-leaf diagrams

Permutations, Combinations & Probability

- Multiplication/addition principles, permutations (nP_r) and combinations (nC_r)
- Conditional probability, independence, tree diagrams, and Venn diagrams
- Discrete random variables: Expectation $E(X)$ and Variance $\text{Var}(X)$
- Binomial distribution: $X \sim B(n, p)$

2. ADVANCED LEVEL (A'LEVEL) FURTHER MATHEMATICS

2.1 PURE MATHEMATICS

Complex Numbers

- Imaginary unit i , complex numbers in $a + bi$ form
- Argand diagrams, modulus $|z|$, and argument $\arg(z)$
- Polar form $r(\cos\theta + i \sin\theta)$ and Exponential form $re^{i\theta}$
- De Moivre's Theorem, expansion of $\cos(nx)$, $\sin(nx)$, and n -th roots of unity
- Loci in the complex plane

Matrix Algebra & Linear Transformations

- Matrix operations, determinants (2×2 and 3×3), and inverse matrices
- System of linear equations and Gaussian elimination
- Eigenvalues, eigenvectors, and characteristic equations
- 2D and 3D geometric linear transformations (rotations, reflections, enlargements, shears)

Advanced Series & Proof

- Proof by Mathematical Induction (summation, divisibility, matrix powers)
- Maclaurin and Taylor series expansions
- Limits using Maclaurin series and L'Hôpital's Rule
- Method of Differences for series summation

Hyperbolic Functions

- Definitions of $\sinh x$, $\cosh x$, $\tanh x$, $\operatorname{sech} x$, $\operatorname{cosech} x$, $\coth x$
- Hyperbolic identities and Osborn's Rule
- Inverse hyperbolic functions and their logarithmic forms
- Calculus involving hyperbolic functions

Differential Equations

- First-order differential equations: Separable variables and Integrating Factor method ($e^{\int P(x)dx}$)
- Second-order homogeneous linear ODEs: $a \frac{d^2y}{dx^2} + b \frac{dy}{dx} + cy = 0$
- Second-order non-homogeneous ODEs: Complementary Function (CF) and Particular Integral (PI)
- Applications to Simple Harmonic Motion (SHM) and damped oscillations

Polar Coordinates & Advanced Calculus

- Converting between polar (r, θ) and Cartesian (x, y) coordinates
- Sketching polar curves and area enclosed using $A = \frac{1}{2} \int r^2 d\theta$
- Integration by parts, reduction formulas, and improper integrals
- Arc length and surface area of revolution (Cartesian, parametric, and polar)

Vectors & 3D Geometry

- 3D vector algebra, dot product, and cross product (vector product)
- Vector equations of lines and planes
- Intersections, angles, and shortest distances (point to line, point to plane, skew lines)

2.2 FURTHER MECHANICS

Motion in a Circle & SHM

- Angular velocity ω , uniform and non-uniform circular motion
- Simple Harmonic Motion differential equation $d^2x/dt^2 = -\omega^2x$
- Displacement, velocity, period, mass-spring systems, and simple pendulums

Variable Force & Collisions

- Motion resisted by speed-dependent forces ($F \propto v$ or $F \propto v^2$)
- Newton's Law of Restitution (e) and direct/oblique impacts in 1D and 2D
- Loss of kinetic energy during collisions

Rigid Body Statics & Dynamics

- Center of mass for standard composite shapes and solids of revolution
- Moment of Inertia (I), Parallel/Perpendicular Axis Theorems
- Rotational dynamics ($\tau = I\alpha$), rotational kinetic energy, and angular momentum

2.3 FURTHER STATISTICS

Continuous Random Variables & Distributions

- Probability Density Function (PDF) $f(x)$ and Cumulative Distribution Function (CDF) $F(x)$
- Mean, variance, median, mode, and percentiles for continuous distributions
- Poisson, Normal, Uniform, and Exponential distributions
- Distribution approximations and continuity corrections

Hypothesis Testing, Chi-Squared & Regression

- Central Limit Theorem (CLT) and Confidence Intervals
- Hypothesis testing (z -tests, single and two-sample t -tests)
- Chi-Squared (χ^2) tests for contingency tables and Goodness-of-Fit
- PMCC, Spearman's rank correlation, and least squares regression lines